

# Rescattering effects for a three-body final state

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**Thanks to:**

Prof. Dr. BERNHARD KETZER, HISKP, Uni Bonn

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# Contents

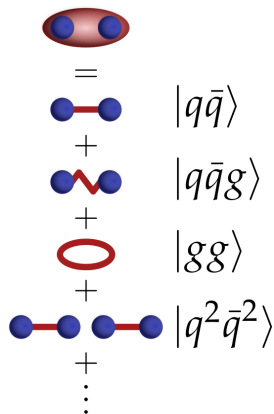
- 1 Motivation
  - Introduction
- 2 Importance of rescattering
  - Final state interaction
- 3 Rescattering equations
  - Unitarity
  - Khuri-Threiman formalism
  - Example
- 4 Conclusion

# Interpretation of resonance

- Beyond bound quark-anti-quark states, other exotic states of QCD are possible
- Possible exotic states are
  - hybrids:  $|q\bar{q}g\rangle$ ,
  - glueballs:  $|gg\dots\rangle$ ,
  - multi-quark states:
    - tetraquarks  $|q\bar{q}q\bar{q}\rangle$ ,
    - hadro-molecules  $|(q\bar{q})(q\bar{q})\rangle$ .
    - ...

Several candidates to be exotic states have been observed

XYZ-states,  $a_1(1420)$ ,  $P_c(4450)$ .

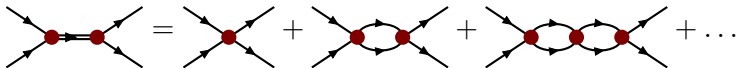


# Two body unitarity

Unitarity = propability conservation.  $\hat{S} \cdot \hat{S}^\dagger = \mathbb{I}$ .

$$\hat{S} = \hat{\mathbb{I}} + i\hat{T} \Rightarrow \hat{T} - \hat{T}^\dagger = i\hat{T}\hat{T}^\dagger, \Rightarrow \Delta T = iT\rho T^*.$$

## Two-body unitarity and resonance



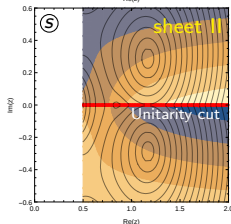
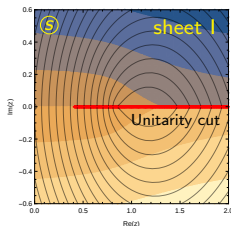
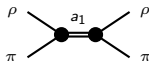
$$T = \frac{g^2}{m^2 - s - im\Gamma} = \frac{1}{\textcolor{red}{K}^{-1} - i\rho} = \textcolor{red}{K} + \textcolor{red}{K} i\rho \textcolor{red}{K} + \textcolor{red}{K} i\rho \textcolor{red}{K} i\rho \textcolor{red}{K} + \dots, \quad \textcolor{red}{K} = \frac{g^2}{m^2 - s}.$$

# Complex structure

- Amplitude  $t(s)$  of scattering process is complex function of  $s$ .
- Every resonance is a pole in the amplitude
- Every one channel for decay produces cut along the real axis

$t^{-1}(s)$  is shown in  $s$ -plane,  
color code is  $\text{Im}[t^{-1}(s)]$ ,  
equipotential lines are  
 $\text{ABS}[t^{-1}(s)]$ .

Two-body resonance

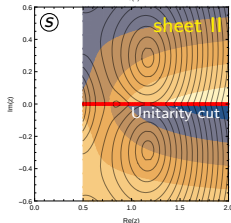
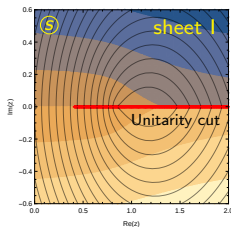
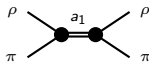


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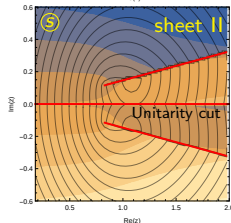
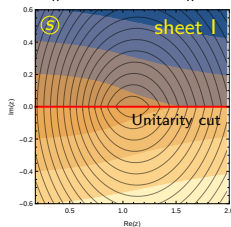
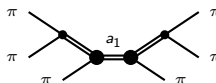
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Two-body resonance



Quasi-two-body resonance

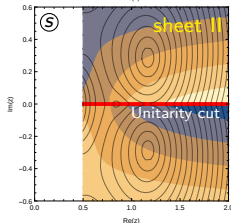
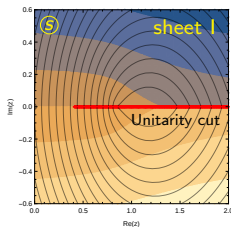
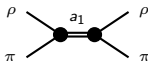


# Complex structure

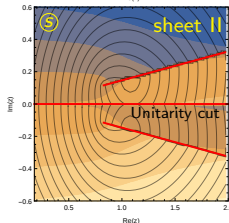
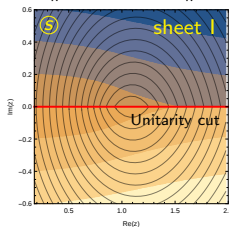
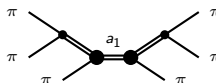
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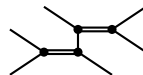
Two-body resonance



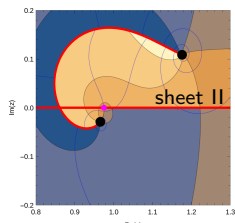
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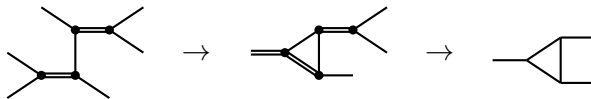
Cross-channel exchange



- Additional structure appears on the second sheet.
- log branching point can be close to the physical region.

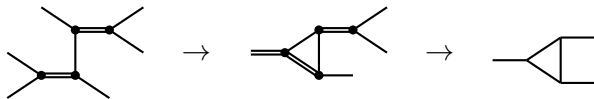


# Triangle diagram



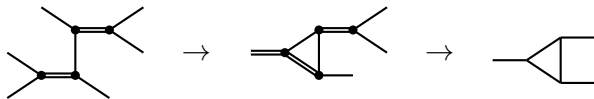


## Triangle diagram



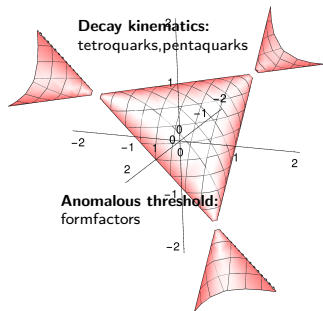
$$\begin{aligned}
& \text{Diagram: } p_0, s_0 \text{ splits into } k_1, m_1^2 \text{ and } k_2, m_2^2, \text{ which then recombine into } p_1, s_1 \text{ and } p_2, s_2. \\
& = g^3 \int \frac{d^4 k_1}{(2\pi)^4 i} \frac{1}{\Delta_1 \Delta_2 \Delta_3} = \frac{g^3}{16\pi^2} \int_0^1 \frac{dx dy dz}{D} \delta(1-x-y-z), \\
& D = x m_1^2 + y m_2^2 + z m_3^2 - x y s_0 - z x s_1 - y z s_2.
\end{aligned}$$

# Triangle diagram



$$\begin{aligned}
 & \text{Triangle diagram with external momenta } p_0, s_0 \text{ and } p_1, s_1, p_2, s_2, \text{ and internal momenta } k_1, m_1^2, k_2, m_2^2, k_3, m_3^2 \\
 &= g^3 \int \frac{d^4 k_1}{(2\pi)^4 i} \frac{1}{\Delta_1 \Delta_2 \Delta_3} = \frac{g^3}{16\pi^2} \int_0^1 \frac{dx dy dz}{D} \delta(1 - x - y - z), \\
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 \end{aligned}$$

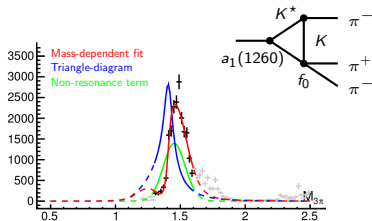
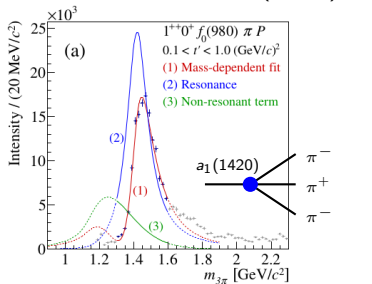
- $\Delta_i = m_i^2 - k_i^2$  is propagators of the particles in the loop,
- Positions of singularities are given by Landau equations. [Landau, Nucl. Phys. **13**, 181 (1959)]
- Landau surface is represented in normalized invariants  $(y_0, y_1, y_2)$ ,  $y_i = \frac{s_i - m_{i1}^2 - m_{i2}^2}{2m_{i1}m_{i2}}$



# Examples

## hypotheses for an explanation of exotics

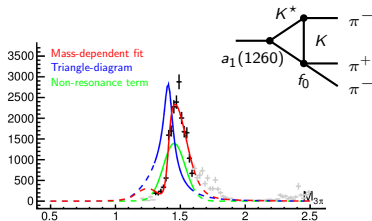
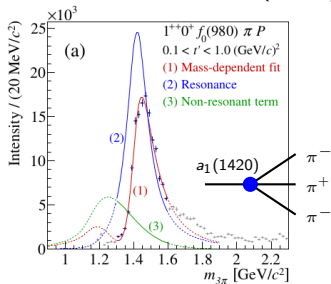
### COMPASS: exotic $a_1(1420)$



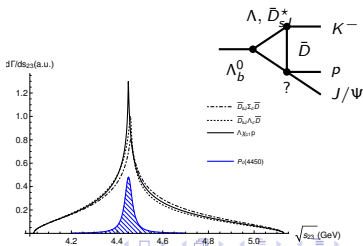
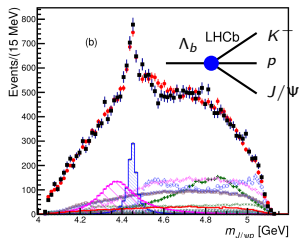
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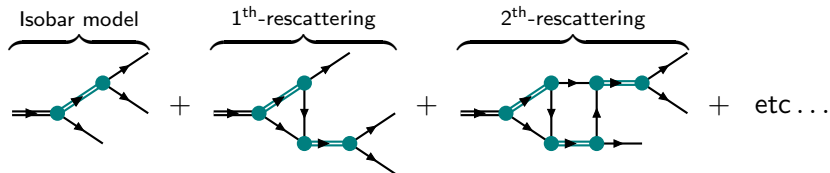
## LHCb: pentaquark $P_c(4450)$



# Rescattering series

How a further rescattering changes an amplitude?

- Isobar model is common tool of the data analysis (PWA at VES, COMPASS, BES, CLEO, LHCb, ...).
- 1<sup>th</sup> order rescattering is triangle diagram. One loop, four integrals can be reduced to one.
- 2<sup>th</sup> is two-loop diagram. Very complicated to evaluate!



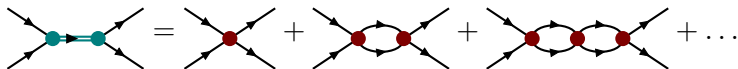
There is alternative method.

## Two body unitarity

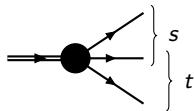
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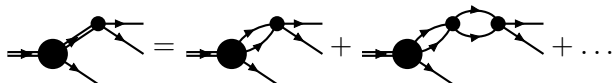
### Two-body unitarity and resonance



$$t = \frac{g^2}{m^2 - s - im\Gamma} = \frac{1}{K^{-1} - i\rho} = K + K i\rho K + K i\rho K i\rho K + \dots, \quad K = \frac{g^2}{m^2 - s}.$$



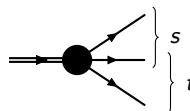
### Subchannel resonances



Complete only if interaction with particle 3 is negligible

# Formalism

## Khuri-Treiman equation



$$\left. \begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \right\}^s = A^{(s)} + A^{(t)} + A^{(u)}, \quad A^{(j)} = \sum_l (2l+1) a_l^{(j)}(s_j) P_l(\cos \theta^{(j)})$$

$$\left. \begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \right\}^t$$

Due to unitarity:  $a_l^{(s)} = \underbrace{t_l^{(s)} c_l^{(s)}}_{\text{Isobar model}} + \underbrace{\frac{t_l^{(s)}}{2\pi} \int \frac{b_l^{(s)} \rho}{s' - s} ds'}_{\text{rescattering corrections}}.$

- $a_l^{(s)}$  is a corrected two-body amplitude,
- $b_l^{(s)}$  is a projection of cross channel waves.
- We get a system of integral equations.



# Input/Output

Model is given by

- Set of waves up to  $L_j$  for every channel  $s, t, u$ .

Example  $\pi^- \pi^+ \pi^-$ : s,t-channel isobars are  $\sigma, f_0(l=0), \rho(l=1)$

- Parameterizations of elastic amplitudes  $t_l^{(j)}$

Example  $\pi^- \pi^+ \pi^-$ :  $(\pi\pi)_S$ -wave,  $(\pi\pi)_P$ -wave from K-function/phase shift



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Result of calculation:

- Every isobar rescatters to all others. A solution of the equations tells **a shape and a strength** of the “induced” waves.

Example  $\pi^- \pi^+ \pi^-$ : Even if  $f_0$  is not coupled to  $a_1$ , it will appear at final state due to rescattering

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Example  $\pi^-\pi^+\pi^-$ : Even if  $f_0$  is not coupled to  $a_1$ , it will appear at final state due to rescattering

- The solution is linear on production constants, thus we can rescatter every wave **independently**.

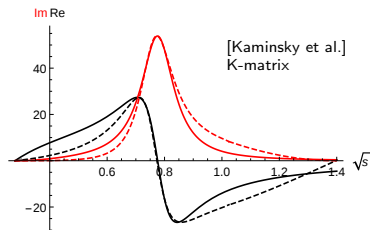
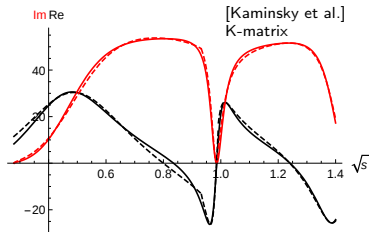
Example  $\pi^-\pi^+\pi^-$ : final shape and strength of  $\rho$ -siobar is

$$A_\rho^{(s)} = (c_\rho a_\rho^{\text{direct}} + c_\sigma a_\rho^{\text{induces}}) P_l(\cos \theta_s)$$

# $(\pi\pi)_S - (\pi\pi)_P$ rescattering

Example:  $0^- \rightarrow \pi^- \pi^- \pi^+$  system:

- $\pi^- \pi^-$  interaction is negligible.
- $\pi^+ \pi^- (\pi\pi)_S$   $l=0$ , i.e.  $f_0(500) - f_0(990) - f_0(1500)$ ,  $(\pi\pi)_P$   $l=1$ , i.e.  $\rho(770)$ .
- Simple pole for  $\rho$ , three K-matrix poles and background for  $(\pi\pi)_S$ .

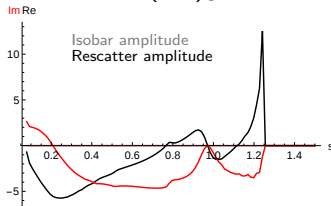
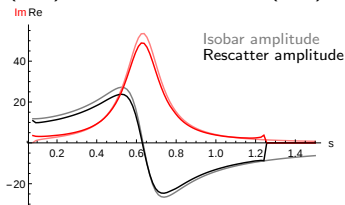


# Representation

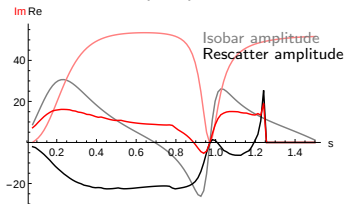
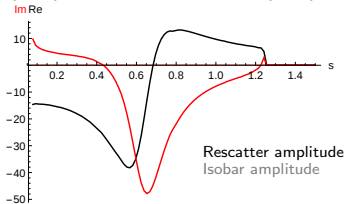
$$A_S^{(s)} = (c_s a_s^{\text{direct}} + c_P a_s^{\text{induces}}) P_0(\cos \theta_s),$$

$$A_P^{(s)} = (c_P a_P^{\text{direct}} + c_S a_P^{\text{induces}}) P_1(\cos \theta_s)$$

- $(\pi\pi)_P$  is a source  $\Rightarrow (\pi\pi)_P$  gets modified,  $(\pi\pi)_S$  is induced.



- $(\pi\pi)_S$  is a source  $\Rightarrow (\pi\pi)_S$  gets modified,  $(\pi\pi)_P$  is induced.



# Conclusion and outlook

## Summary

- Isobar model is a rough approximation, unitarity is not satisfied.
- Rescattering correction can be significant.
- Khuri-Treiman equation gives a tool to deal with a complete series of rescattering.
- KT-PWA analysis is promising extension of PWA.

## Prospectives:

- Generalisation of the formalism of the decay of a particle with a spin.
- Implementation of formalism for coupled channels.